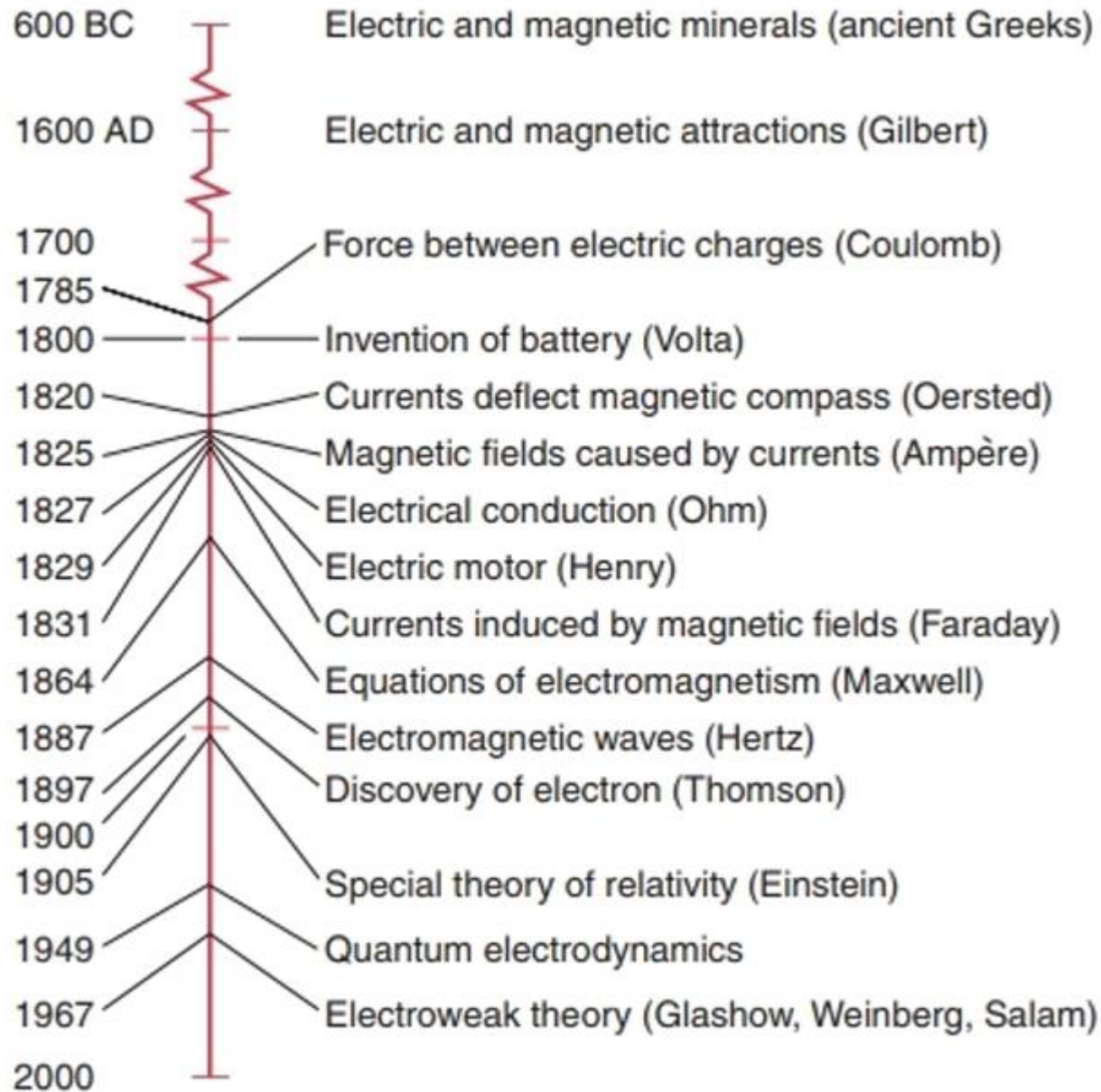


Week 1

Electrostatic



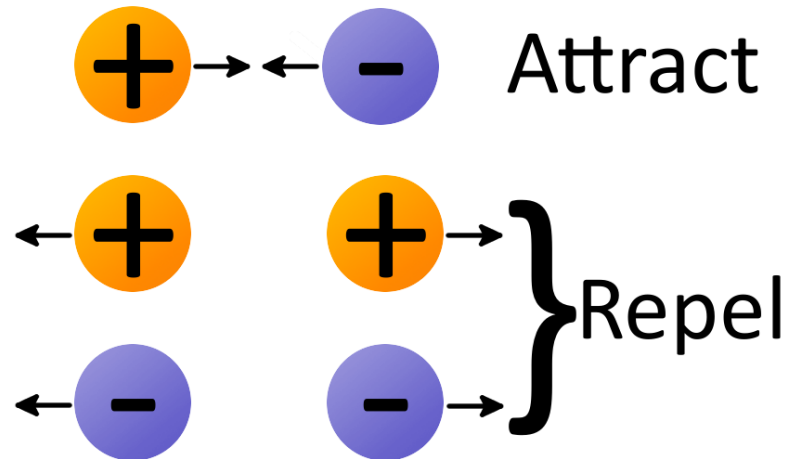
Muhammad Daniyal Chundrigar
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Time line of major developments in electromagnetism.

Electric charge: An intrinsic property of fundamental particles which takes parts in conduction process is called electric charge. Either they repel or attract each other. Kinds of charges

There are two types of charge, **positive and negative**, charge on proton is positive and charge on electron is negative.



Coulomb's Law for static charges Vector Form

The force of attraction or repulsion b/w two charges is directly proportional to the product of the magnitude of charges and inversely proportional to the square of distance b/w them.

$$F \propto q_1 q_2$$

$$F \propto \frac{1}{r^2}$$

$$F \propto \frac{q_1 q_2}{r^2}$$

$$F = \frac{K q_1 q_2}{r^2}$$

K (*Coulomb's constant*)

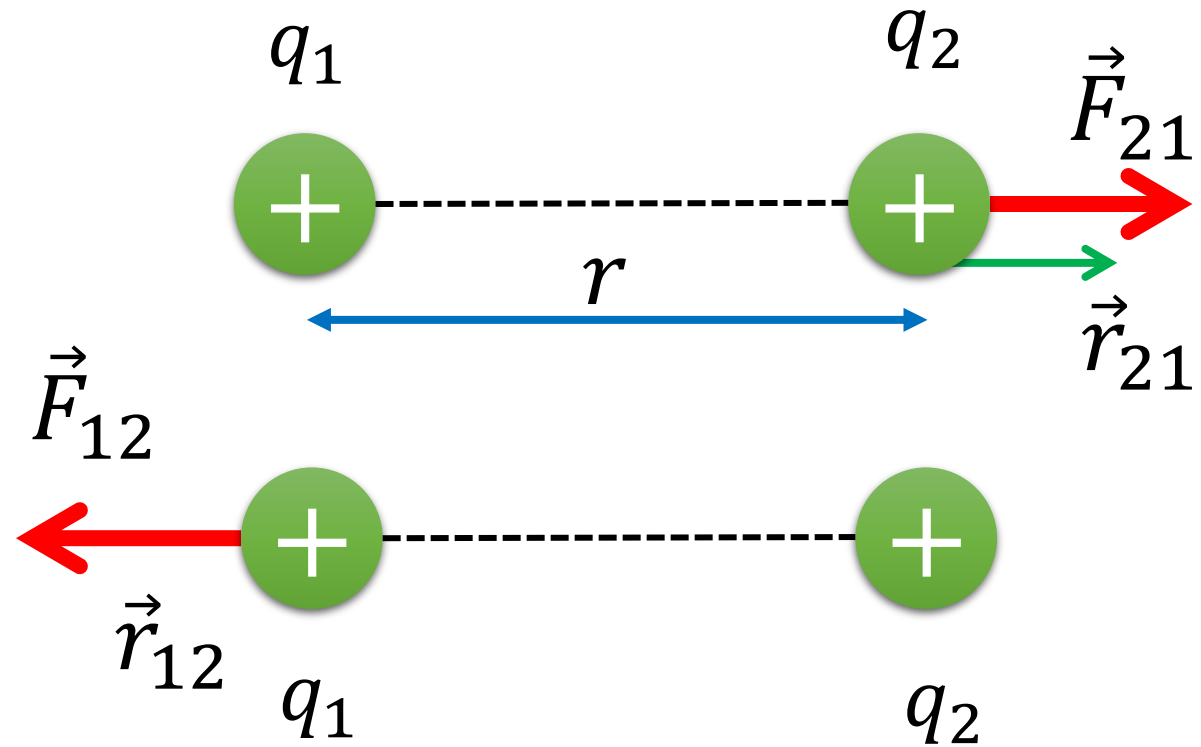
$$K = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

$$K = \frac{1}{4\pi\epsilon_0}$$

Coulomb's Law for static charges Vector Form

The force of attraction or repulsion b/w two charges is directly proportional to the product of the magnitude of charges and inversely proportional to the square of distance b/w them.

$$\vec{F}_{21} = \frac{kq_1q_2}{r^2} \hat{r}_{21}$$





Sample Problem 1.1: The average distance r between the electron and the proton in the hydrogen atom is $5.3 \times 10^{-11}m$. (a) What is the magnitude of the average electrostatic force that acts between these two particles? (b) What is the magnitude of the average gravitational force that acts between these particles?



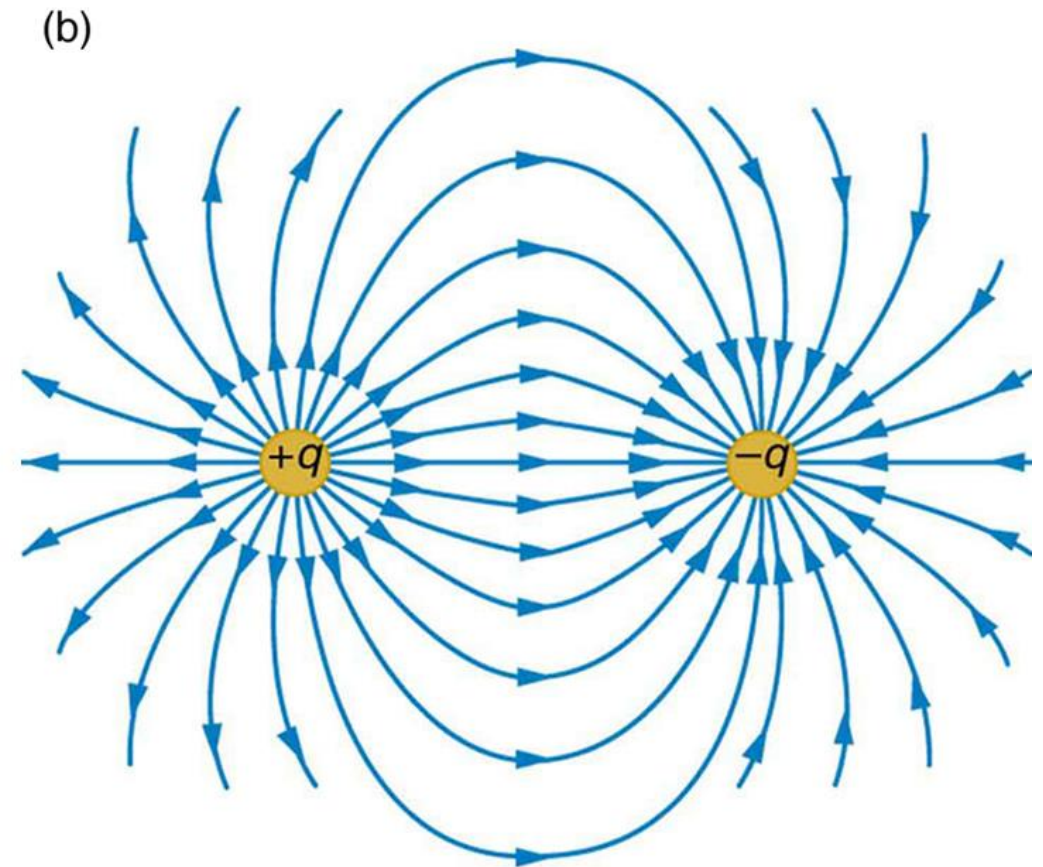
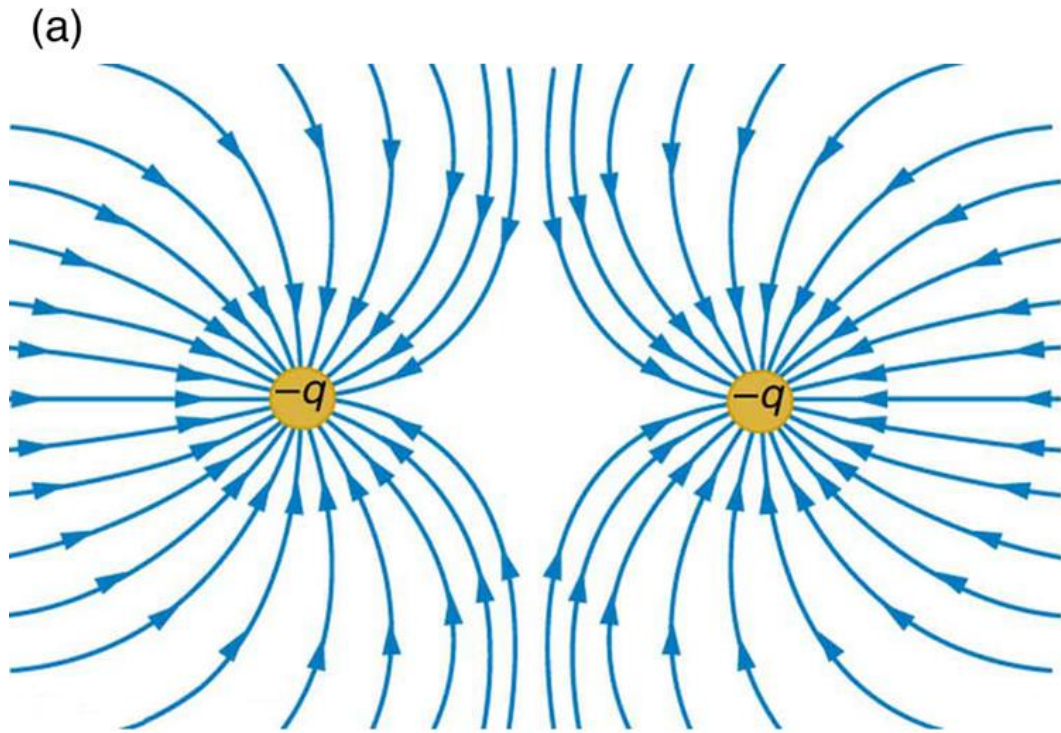
Sample Problem 1.1: The average distance r between the electron and the proton in the hydrogen atom is $5.3 \times 10^{-11}m$. (a) What is the magnitude of the average electrostatic force that acts between these two particles? (b) What is the magnitude of the average gravitational force that acts between these particles?

Hint:

The mass of a proton is
approximately (m) $1.67 \times 10^{-27}kg$

Universal gravitational Constant
(G) $6.67 \times 10^{-11}Nm^2/kg^2$

Electric field: “The space or region around a charge within which another charge experience a force is called electric field”. Michael Faraday introduced the concept of electric field.





Electric field intensity due to a point charge q: The force experienced by test charge q_0 placed in the field of charge q is calculated by Colomb law and electric field intensity is calculated by putting in formula of field intensity.

$$\vec{E} = \frac{\vec{F}}{q_0} \text{ ---- (1)}$$

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} \hat{r} \text{ ---- (2) Put this value in equation (1)}$$

$$\vec{E} = \frac{1}{q_0} \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} \hat{r} \quad \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \text{ Vector Form} \quad E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \text{ Scalar Form}$$

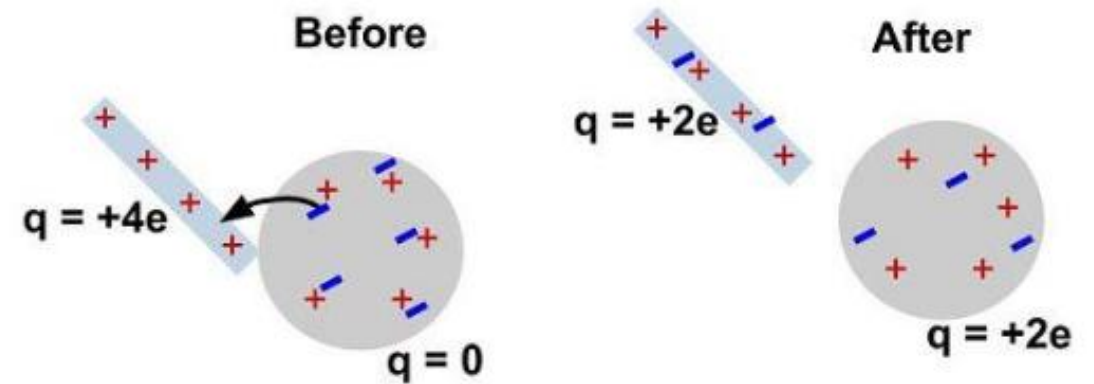
Electric field intensity depends upon a) Magnitude of charge b) medium c) distance from charge

Conservation of charge and charge quantization

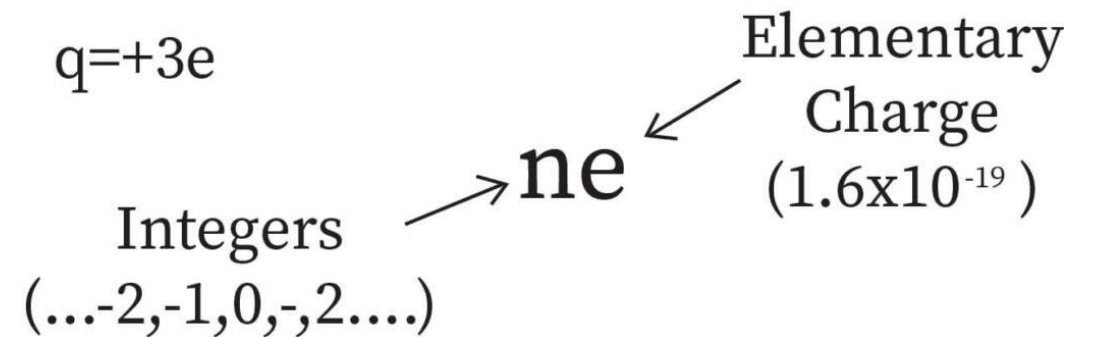


In an isolated system, the total charge (sum of positive and negative) remains constant whatever change takes place in that system.

The charge is quantized: Charge on anybody always exists in integral multiples of a fundamental unit of electric charge. This unit is equal to the magnitude of charge on an electron ($1e = 1.6 \times 10^{-19}$ coulomb). So charge on anybody $Q = \pm ne$, where n is an integer and e is the charge of the electron.



Quantization of Electric Charge



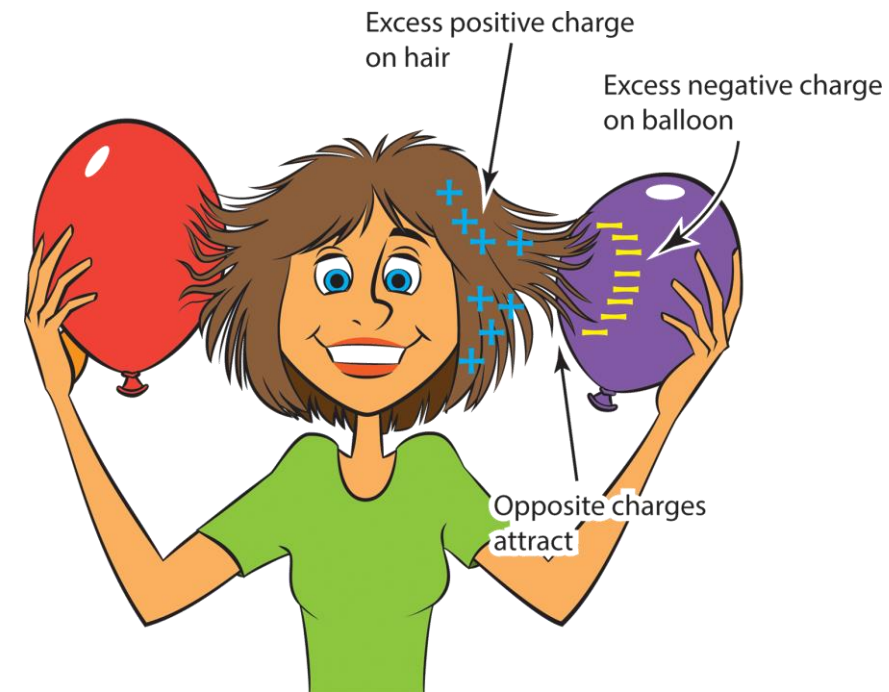
Introductory Note

These all are links for just motivation. My lecture was already enough for understanding the concepts. I put a lot of effort into it. However, some points need more attention or repetition, so you can check these additional resources.

I played with several simulations in class, which I personally love, and I've collected all these lectures from good and reliable sources such as Khan Academy, Professor Dave, and others.

Lectures:

1. <https://www.youtube.com/watch?v=MsINeZMeTMQ&t=299s>
2. https://www.youtube.com/watch?v=VFbyDCG_j18
3. <https://www.youtube.com/watch?v=5NiB6LXF13Y>
4. <https://www.youtube.com/watch?v=m5VbK66a254>
5. <https://www.youtube.com/shorts/sXmcgTOkmyA>
6. <https://www.youtube.com/watch?v=Y0Ai704mkIY>
7. <https://www.youtube.com/watch?v=rYjo774UpHI>
8. <https://www.youtube.com/watch?v=NrVMHESQgi4>



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Experiments:

1. <https://www.youtube.com/watch?v=9mFIElwuctI>
2. https://www.youtube.com/watch?v=D5pJKu_QjRo

Simulations:

1. https://phet.colorado.edu/sims/html/charges-and-fields/latest/charges-and-fields_en.html
2. https://phet.colorado.edu/sims/html/balloons-and-static-electricity/latest/balloons-and-static-electricity_all.html
3. <https://phet.colorado.edu/en/simulations/coulombs-law>

